

IERG3840 Web Application Development Project (2025-26 Term 1)

Lab 4 Specification

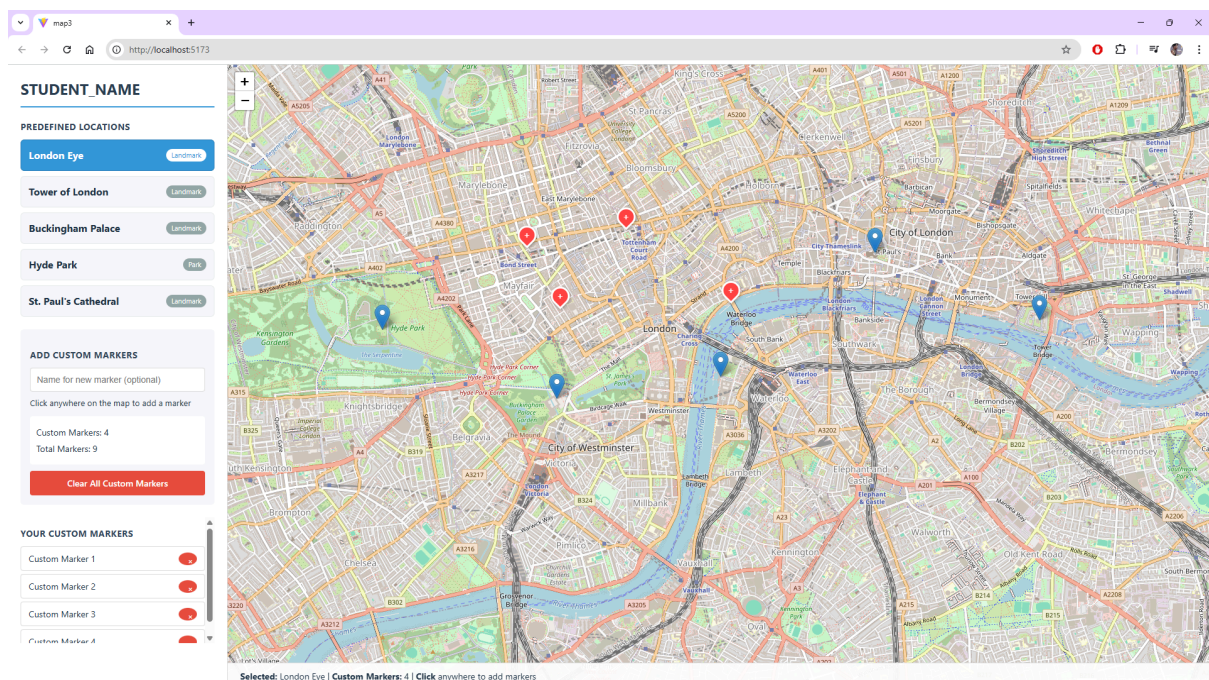
Lab 4 is to design a React Leaflet map application with or without the support of Node.JS server in your laptop. After understanding the demonstration of map applications in the lecture note, you are going to implement your advanced map application. It will take 15% of the course total.

Part 1: Demonstration (7% + Bonus 1%)

Date: 30th Oct, 2025 (Thur)

Time and Venue: In Class

You need to bring your own laptop for demonstration



In your demonstration, you need to demonstrate the following features:

- Show your full name (0.5%)
- A basic map application is presented. The user can zoom in, zoom out and drag to navigate. (0.5%)
- The map's default starting location is determined by the last digit of your student ID. Upon starting, the map will always be centered on this location. All locations referenced are landmarks at CUHK, which you must identify the position on your own. (1%)

Last digit of your student ID	Student ID example	Default starting location of your map
0	1155000000	Lake Ad Excellentiam
1	1155000001	Pavilion of Harmony
2	1155000002	Chinese Pillars
3	1155000003	Chung Chi Gate
4	1155000004	Science Centre
5	1155000005	University Library
6	1155000006	The Beacon
7	1155000007	Glorious United Man
8	1155000008	Shaw Terrace
9	1155000009	Chung Chi College Chapel

- The map features different markers (custom markers and shop markers). Clicking a marker will open a map popup box displaying its details, including the marker type and its location. (1%)
- Users can create custom markers by clicking on the map. These markers are stored in a list that is visible to the user. A “Clear Markers” button allows the user to remove all custom markers at once. (1%)
- Clicking the “My Location” button will prompt the browser to obtain the laptop's current location. The map will then center on this location, create a custom marker for it, and add the marker to the list mentioned above. (1%)

- You must find three different locations for your assigned shop, which is based on the second last digit of your student ID. These shop locations are displayed in a separate list and are marked on the map with the shop icon. These shop markers are distinct from the custom markers and should not be included in the list of custom markers. All shop markers are loaded onto the map when the application starts. Clicking a shop in the list will re-center the map on that shop's location. (2%)

Remarks: The shop locations must be authentic.

Second last digit of your student ID	Student ID example	You need to find three different locations for your assigned shop
0	11550000 00	7-11
1	11550000 10	Circle K
2	11550000 20	Fusion
3	11550000 30	Market Place
4	11550000 40	Watsons
5	11550000 50	McDonald's
6	11550000 60	KFC
7	11550000 70	Pizza Hut
8	11550000 80	MOS Burger
9	11550000 90	Saizeriya

- [This task requires self-learning and is tedious] Session or cookie (Bonus 1%):
 - If you use sessions: The Node.JS server will store the list of custom markers in a session. This allows the markers to persist after restarting the application.
 - If you use cookies: The Vite and React app will store the list of custom markers in the user's browser cookies. This also allows the markers to persist after restarting the application.

Part 2: Question and Answer (5%)

You will be asked one set of problems out of the five sets below randomly after completing the demonstration. You need to answer the problem set verbally.

You can prepare a cheat sheet for the Q&A session, however, you cannot search online or search from the lecture notes during the Q&A session.

There are five points in each set.

Set 1:

- (a) Suggest two major differences between Git and GitHub. (2%)
- (b) Is the git commit message unique? Explain briefly. (1%)
- (c) Suggest two best practices for git commit message. Explain briefly. (2%)

Set 2:

- (a) What is a 'branch' in git? (1%)
- (b) What is a 'conflict' in git? (1%)
- (c) Suggest the steps in resolving 'conflict' in git? (3%)

Set 3:

- (a) What is OAuth? (2%)
- (b) How does it work? Use Google OAuth 2.0 for web server applications as an example. (3%)

Set 4:

- (a) In Google OAuth 2.0, what is Google client ID? (1%)
- (b) In Google OAuth 2.0, what is Google client secret? (1%)
- (c) Explain the steps on how we can obtain the Google client ID and Google client secret. (3%)

Set 5:

- (a) Suggest two advantages of building a login and registration system in lab 3, over using the Google OAuth 2.0. (2%)
- (b) How can Google OAuth 2.0 return whether the user is authorized or not to the application? (1%)
- (c) What are scopes in OAuth 2.0? (2%)

Part 3: Short Report (3%)

Use the template, fill in all the necessary information and submit the **PDF** format on or before 31st Oct, 2025 (Fri). Without demonstration, no scores will be given for the submitted report.

Late Policy

The following behaviours are regarded as late:

- Demonstration and Q&A are not performed on the specified date without justification
- Short report is submitted after the deadline

Late penalties will be counted on the late parts only.

- Demonstration and Q&A: 30% off for late within one week
- Report: 30% off for late within one week
- No scores after one week

< End of Specification >